Frequency tables

1 A gymnast received the following scores over 12 rounds of a competition:

Calculate the missing frequency to find how many times the gymnast scored 8.

Score	5	6	7	8	9
Frequency	2	1	1		1

2 Bill asks nine of his friends which pet they would like and writes down their responses below:

Dog, Fish, Cat, Cat, Fish, Dog, Fish, Fish, Dog

Construct a frequency table for these responses.

3 Brad stands at the school gate and tallies the hair colour of students as they enter the school. His results are shown in the table:

a Construct a frequency table from Brad's tally.

b Find the total number of students Brad tallied.

c What fraction of students had red hair?

Hair Colour	Tally
Brown	𝍸 𝍸 𝍸 𝍸 𝍷
Blonde	𝍸 𝍷𝍷𝍷
Black	𝍸 𝍸 𝍷𝍷𝍷
Red	𝍷𝍷

4 The following table shows the number of trains arriving either on time or late, at Wallarobba station during one week:

a How many trains arrived on time on Thursday?

b How many trains arrived late on Wednesday?

c What fraction of the total trains arriving during the week, arrived on Friday?

d What fraction of the trains arriving on Sunday were on time?

Day	Trains on time	Trains arriving late
Monday	14	6
Tuesday	16	4
Wednesday	18	2
Thursday	15	5
Friday	17	3
Saturday	6	6
Sunday	5	3

5 Fifteen airline passengers weigh their lug[redacted] The weights in kilograms are listed below:

$$17, 20, 19, 19, 20, 21, 20, 22, 20, 19, 17, 16, 20, 20, 16$$

a Construct a frequency table for this data.

b How many passengers had luggage that weighed more than 20 kilograms?

c The airline charges each passenger $\$3$ for every kilogram above $20\ \text{kg}$. Calculate the total charge for overweight luggage for this group of passengers.

6 A school nurse is investigating the number of sick days taken by students in a class, and constructs the following table:

a Construct a frequency table for the number of sick days.

b State the number of students who had no sick days.

c State the most common number of sick days for the class.

Student	Sick days
Sally	0
Lachlan	2
Tara	1
Xavier	4
Luigi	2
Nadia	4
Lisa	0
Dave	1
Bill	3
Danielle	4

7 Find the mode(s) of the following sets of scores:

a $12, 12, 13, 11, 13, 9, 13, 14, 9$

b $14, 2, 5, 2, 2, 5, 5, 7, 12, 13, 13, 14$

c

Score	21	22	23	24	25	26
Frequency	31	37	39	33	38	34

8 Yvonne asks a number of her friends what their favourite colour is, and writes down their reponses below:

Blue, Pink, Blue, Yellow, Green, Pink, Pink, Yellow,

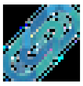
Green, Blue, Yellow, Pink, Yellow, Pink, Pink

a Construct a frequency table.

b How many friends did Yvonne survey?

c Which colour is the mode?

9 Mrs Tran asks her students how many bedrooms they have in their house and writes their responses below:



$$3, 3, 4, 4, 3, 3, 4, 5, 2, 2, 4, 4$$

a Complete the frequency table.

b How many students did Mrs Tran survey?

c State the modal number of bedrooms.

Number of Bedrooms	Number of Students
1	
2	
3	
4	
5	

10 Ms. Lee bought pizza for her class. She told her students they could eat as much as they wanted in two minutes. She counted up how many slices each student ate in that time and wrote her results below:

$$2,\ 2,\ 2,\ 3,\ 6,\ 0,\ 2,\ 3,\ 4,\ 2,\ 1,\ 1,\ 2,\ 2,\ 6$$

a Complete the frequency table for this data.

b How many students are in the class?

c State the modal number of slices eaten.

Slices eaten	Number of Students
0	
1	
2	
3	
4	
5	
6	

Grouped data

11 As part of a fuel watch initiative, the price of petrol at a service station was recorded each day for 30 days, and displayed in the frequency table:

a What was the maximum price that could have occurred in the 30 day period?

b Find the number of days where the price was above 139.9 cents.

Price (in cents per litre)	Frequency
$130.0 - 134.9$	4
$135.0 - 139.9$	9
$140.0 - 144.9$	12
$145.0 - 149.9$	5

12 The frequency table shows the resting heart rates of a number of people taking part in a study:

Heart Rate	Frequency
$30-39$	20
$40-49$	22
$50-59$	28
$60-69$	37

a Find the number of people taking part in the study.

b How many people had a resting heart rate between 40 and 59 beats/minute?

c How many people had a resting heart rate of below 50 beats/minute?

13 A group of 30 people was surveyed about how many video games they had played in the past month. The results are shown in the given table:

Number of video games played	Frequency
$0-4$	5
$5-9$	12
$10-14$	9
$15-19$	4

a Find the number of people who played 10 or more video games.

b Find the number of people who played less than 5 games.

c Find the number of people who played more than 4, but less than 15 video games.

d State the modal class for the data.

14 The heights of 200 selected rugby players were measured and results recorded in the following table:

Height (cm)	Frequency
$170-174$	8
$175-179$	26
$180-184$	61
$185-189$	55
$190-194$	41
$195-199$	7
$200-204$	2

a If the average height of an Australian man is 175 cm, find the number of rugby players that are shorter than average.

b Find the number of rugby players with heights greater than 189 cm.

15 The ages of golfers in a competition are listed below:

$63, 69, 62, 61, 61, 68, 66, 63, 67, 66, 62, 66, 61, 69, 59, 63, 69, 63, 57, 60, 57, 67, 59, 62, 62, 57$

Complete the frequency table for this data.

Age	Frequency
$57-61$	
$62-66$	
$67-71$	
Total	

16 A tree farmer measured the height of newly planted saplings to the nearest centimetre. The results are shown below:

$$16,\ 17,\ 27,\ 36,\ 16,\ 27,\ 22,\ 37,\ 17,\ 17,$$
$$26,\ 22,\ 37,\ 27,\ 22,\ 21,\ 26,\ 27,\ 37,\ 31,$$
$$27,\ 37,\ 27,\ 21,\ 22,\ 32,\ 32,\ 36,\ 17,\ 32,$$
$$31,\ 37,\ 27,\ 26,\ 37,\ 32,\ 36,\ 27,\ 26$$

a Complete the frequency table.

b State the range.

c Find the number of saplings that were measured.

d In which height interval does the mode lie?

e Find the number of saplings that have a height lower than 25 cm.

Height (cm)	Frequency
$15-19$	
$20-24$	
$25-29$	
$30-34$	
$35-39$	

17 A group of high school students wanted to convince their principal that the school needed airconditioning. They measured the temperature (in $^\circ\mathrm{C}$) in a classroom at 1 pm every day during February and recorded the following results:

$$35,\ 26,\ 32,\ 29,\ 29,\ 32,\ 26,\ 29,\ 35,\ 23,\ 23,\ 32,\ 35,\ 26,$$
$$26,\ 23,\ 26,\ 29,\ 32,\ 35,\ 23,\ 26,\ 29,\ 29,\ 29,\ 32,\ 23,\ 29$$

a Complete the frequency table.

b Which temperature group is the highest recorded temperature?

c How many days in February had a temperature higher than $30^\circ\mathrm{C}$?

d Hence calculate the number of days that had a temperature lower than or equal to $30^\circ\mathrm{C}$.

Temperature in $^\circ\mathrm{C}$	Frequency
$22-24$	
$25-27$	
$28-30$	
$31-33$	
$34-36$	

18 Neil tracks how many views his website receives each week for ten weeks. The results are given in the following table:

Week	1	2	3	4	5	6	7	8	9	10
Views	601	612	618	620	636	643	644	659	670	691

Construct a frequency table for this data set by grouping the data into sets of twenty, with the first group being $600-619$.

19 Consider the data below:

$$48, 31, 85, 46, 56, 33, 66, 48, 86, 74, 64, 63, 50,$$
$$52, 62, 67, 33, 66, 81, 40, 58, 63, 68, 58, 78$$

Construct a frequency table for this data by grouping the data into sets of ten, with the first group being
$30 - 39$.

20 A PE teacher asked her students to do as many push-ups as they could, and recorded the number of push-ups each student completed below:

$$38, 31, 12, 14, 29, 7, 21, 19, 46, 13, 23, 3, 13, 17, 4, 6, 11, 24, 5, 20$$

a Construct a frequency table for this data set by grouping data into sets of ten, with the first group being $0 - 9$.

b Find the number of students in the class.

c How many students could perform at least 30 pushups?

d How many students could not perform at least 40 pushups?